

Hardware Documentation

2.8inch IPS ESP32-S3 (ES3C28P & ES3N28P) Display Module

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1. Hardware Overview

The display module integrates a 2.8-inch IPS LCD, capacitive touch, ESP32-S3 main control, audio processing, and various peripheral interfaces. Below is the mapping of the physical components on the PCB.

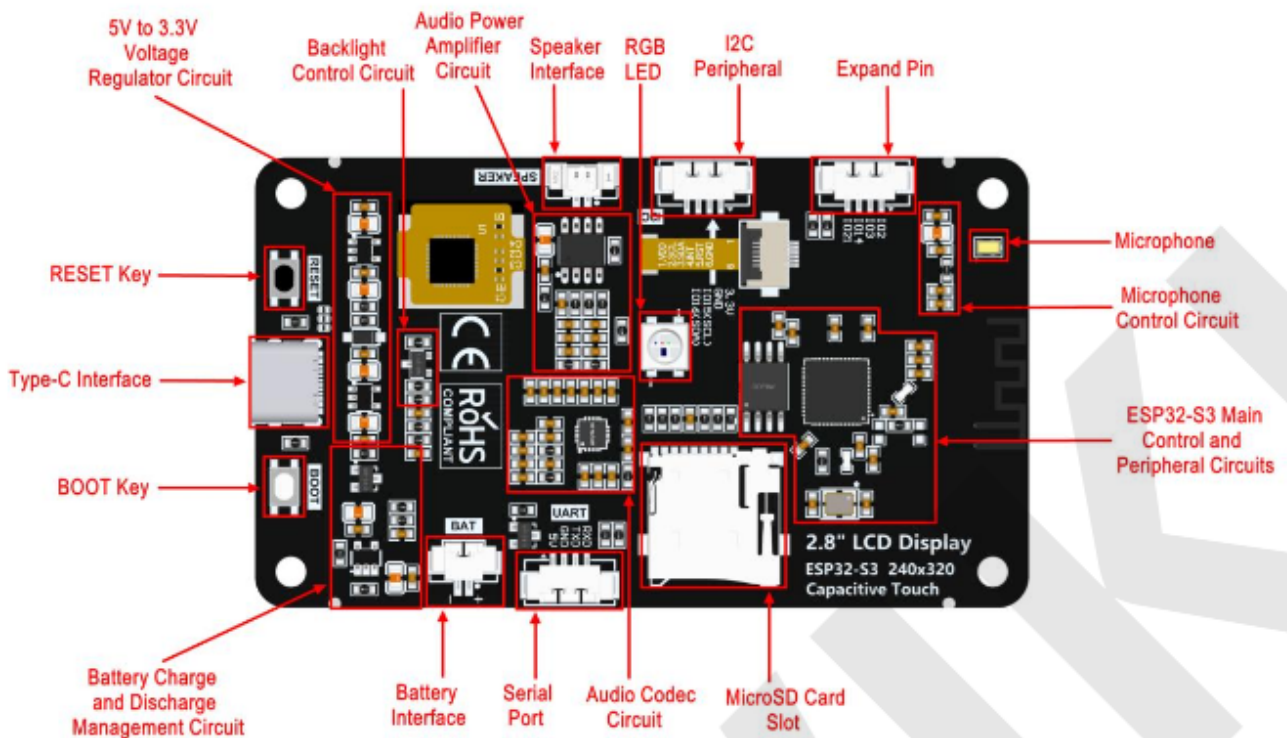
Product Picture



ES3C28P-TopView



ES3C28P-BottomView



2. Display & Touch

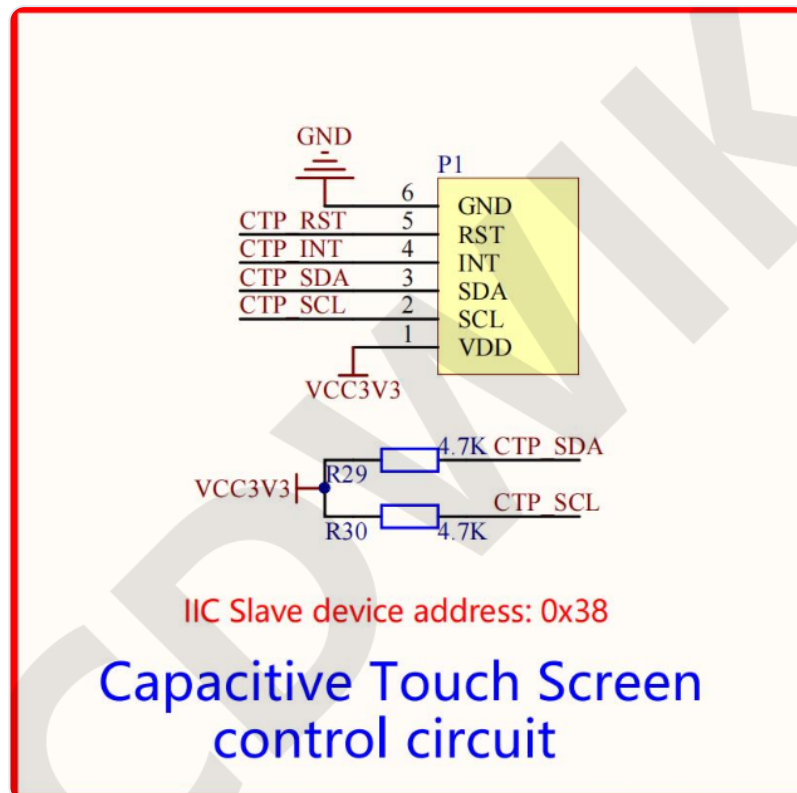
LCD Screen

The display is a 2.8-inch IPS screen driven by the **ILI9341V** IC with a resolution of 240x320. It connects to the ESP32-S3 using a 4-wire SPI communication interface. The ILI9341 uses 16 bits (RGB565) to control pixel display, supporting up to 65K colors.

Note: Refer to the ILI9341V datasheet for exact SPI timing diagrams and format structures.

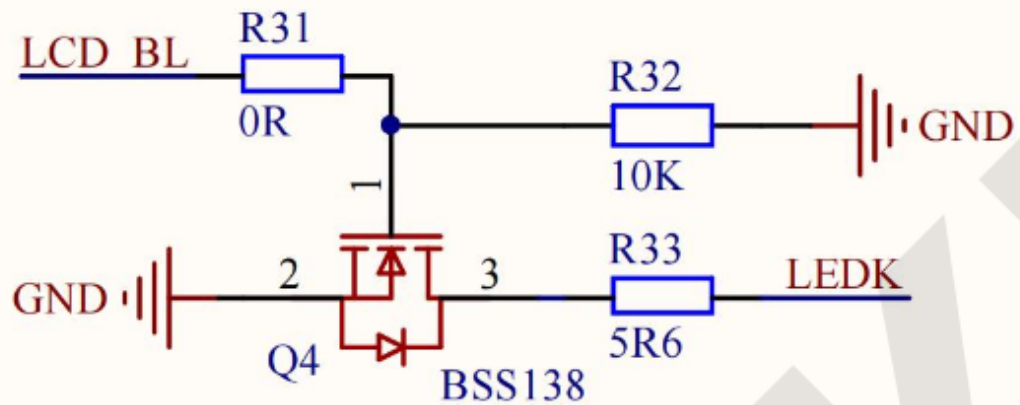
Capacitive Touch Screen

The touch interface uses the **FT6336G** control IC and communicates with the ESP32-S3 via the I2C bus (slave address: **0x38**). It connects through a 0.5mm pitch FPC socket using four pins: **CTP_RST**, **CTP_INT**, **CTP_SDA**, and **CTP_SCL**.



LCD Backlight Control

Driven by a **BSS138** N-channel MOSFET. A high level on the **LCD_BL** pin conducts the drain/source, illuminating the screen. It supports PWM signal input for brightness adjustment.

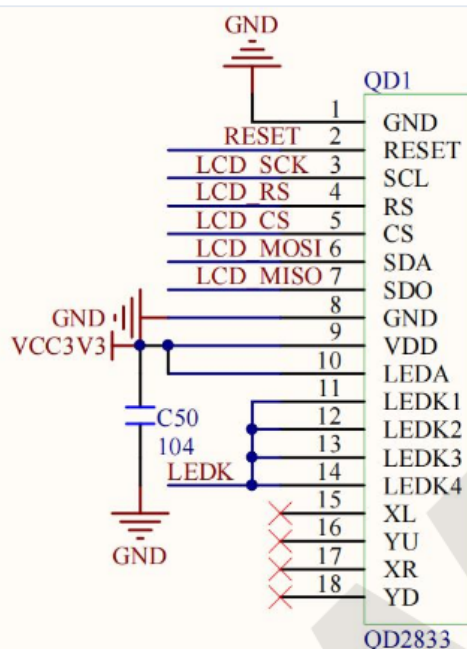


LCD screen backlight control circuit

High level lighting backlight

18P LCD Panel Welding Interface

An 18-pin 0.8mm pitch interface that carries the SPI signals, screen voltage pins, backlight circuit pins, and touch signals.

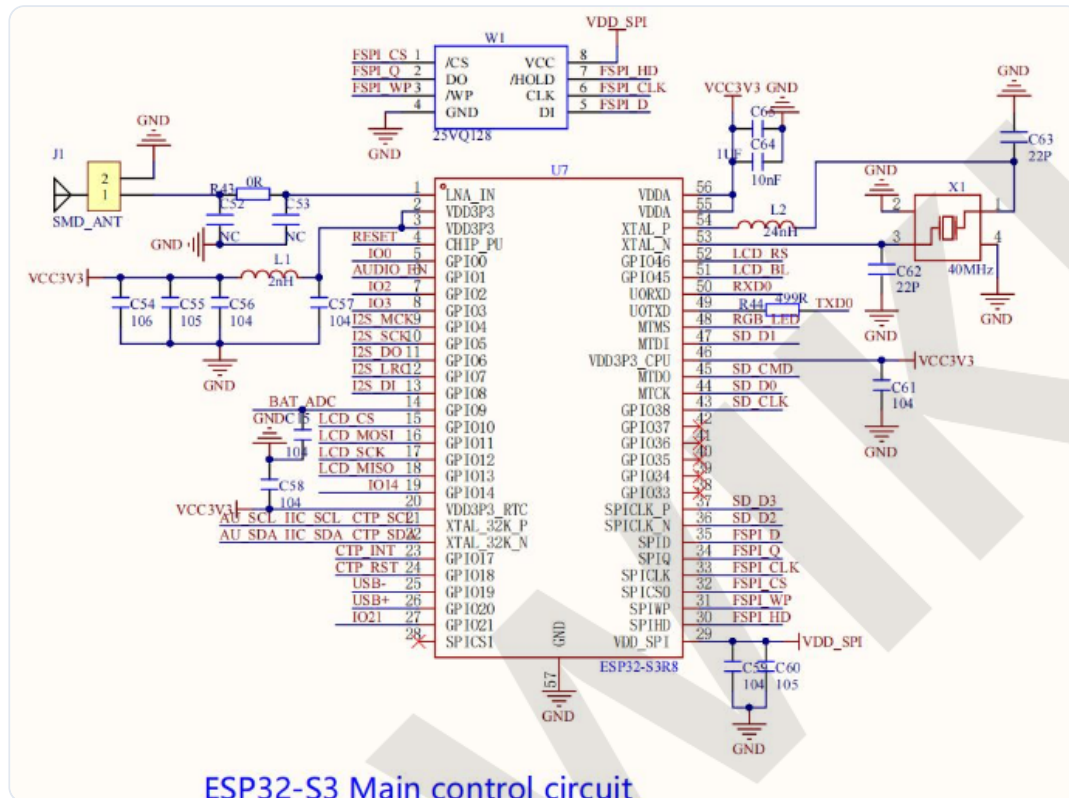


18P LCD panel wire welding interface

3. Core Controller

The main control chip is the **ESP32-S3R8**, featuring an Xtensa dual-core 32-bit LX7 microprocessor (up to 240MHz). It includes 8MB of internal OPI PSRAM and connects to an external 16MB QSPI Flash.

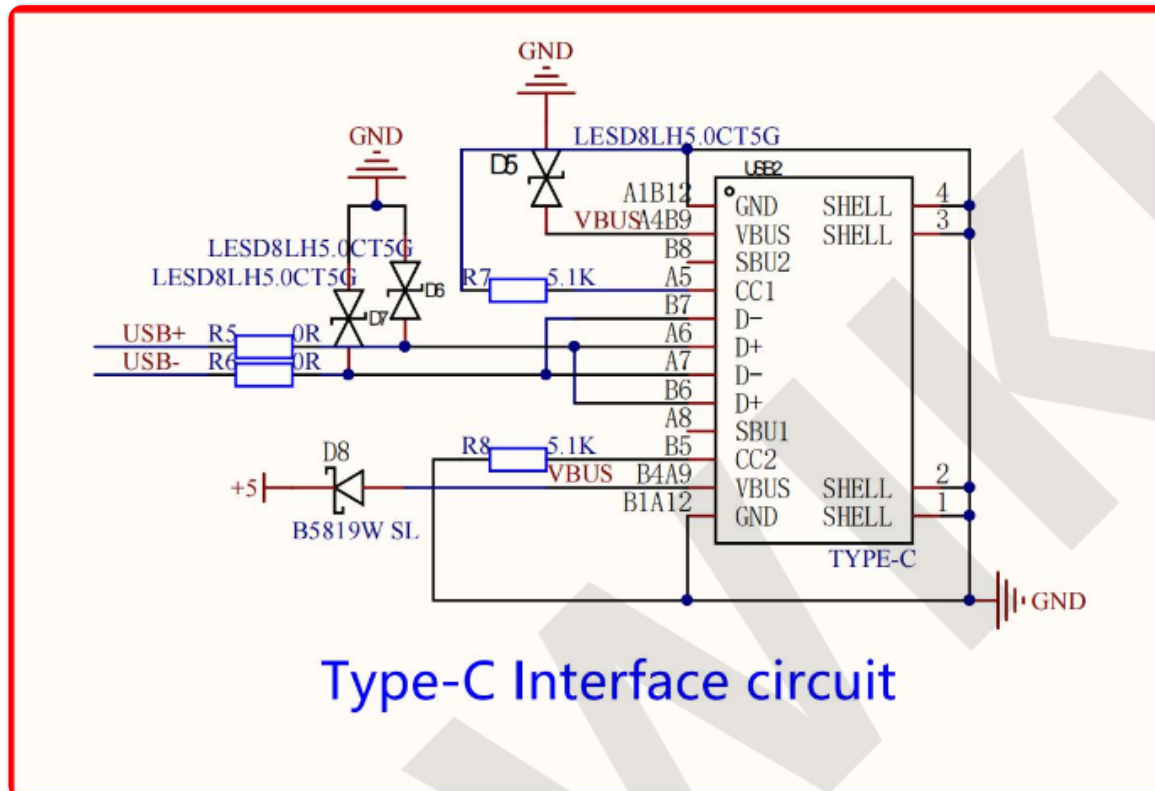
- Supports 2.4GHz Wi-Fi and Bluetooth V5.0.
- Equipped with an onboard PCB antenna circuit.
- Utilizes a 40MHz passive crystal oscillator.
- Includes a CLC filtering circuit for analog power.



4. Power Management

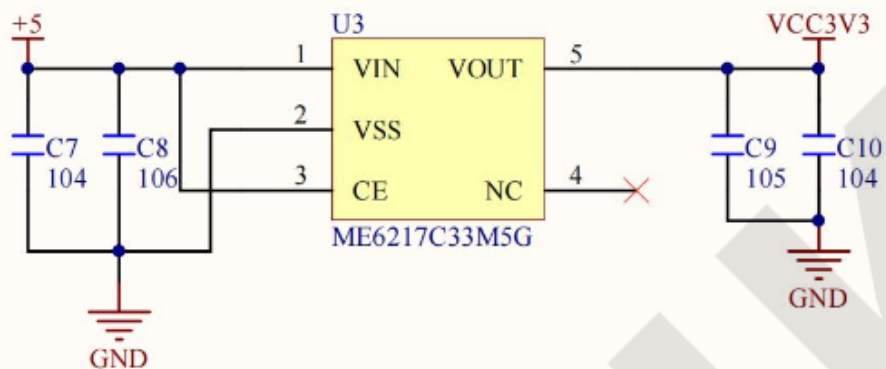
Type-C Interface

Provides main power (5V), program downloading, and serial communication. Features a Schottky diode to prevent current reversal and static surge protection diodes.

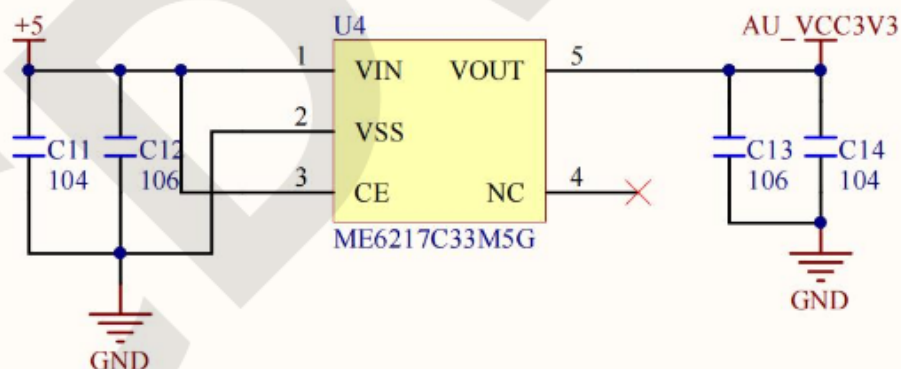


5V to 3.3V Voltage Regulators

Two separate **ME6217C33M5G** LDOs are used (max 800mA output) to isolate power paths: one exclusively for the audio circuits and one for the rest of the board.



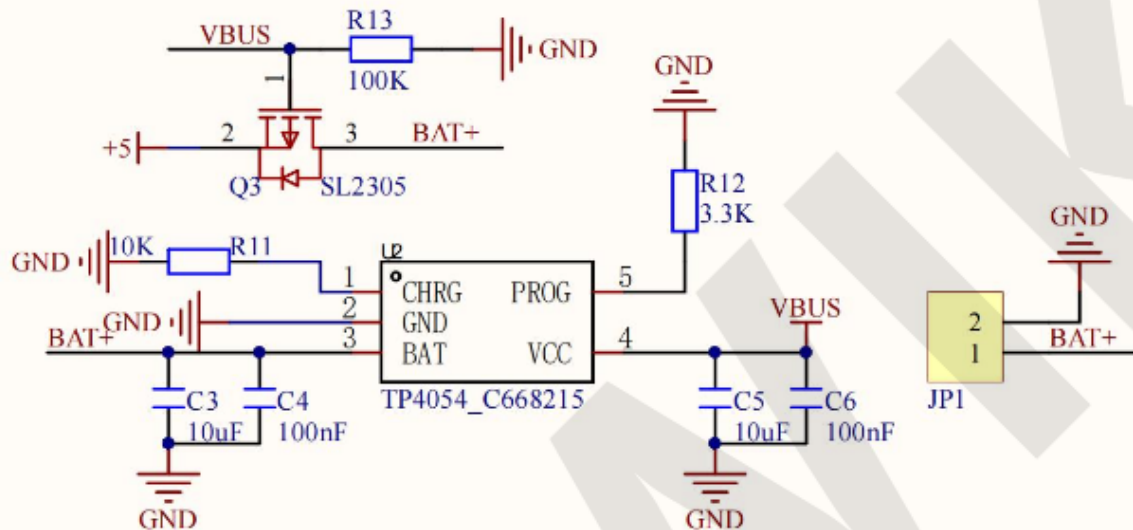
5V to 3.3V voltage regulator circuit



Audio 5V to 3.3V voltage regulator circuit

Battery Management (Charge & Discharge)

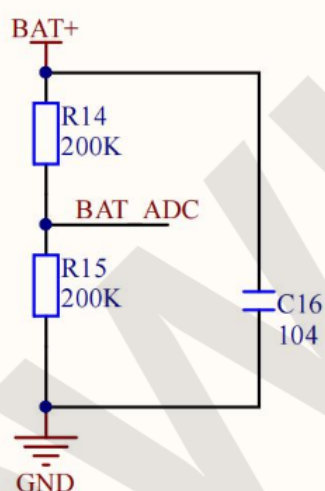
Controlled by the **TP4054** IC. A 2-pin 1.25mm pitch interface allows connecting a battery. The TP4054 limits charging current via a resistor (up to 500mA). A P-channel MOSFET (Q3) cuts off battery power when 5V Type-C power is present.



Battery charge and discharge management circuit

Battery Level Detection

Since the battery saturation voltage (4.2V) exceeds the ESP32-S3's ADC maximum (3.3V), a voltage divider (two 200K resistors) scales the voltage down. The ADC reads this value, which must be multiplied by 2 in software to get the true battery voltage.

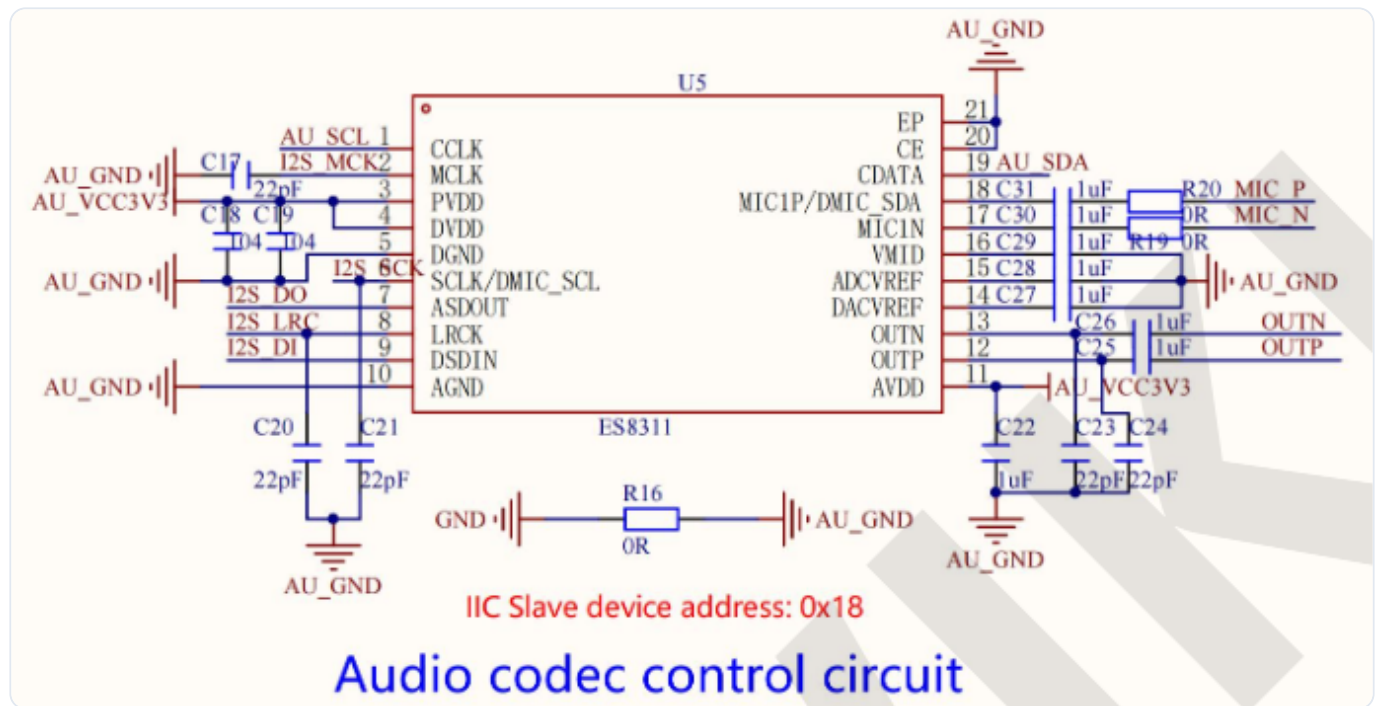


Battery level detection circuit

5. Audio Subsystem

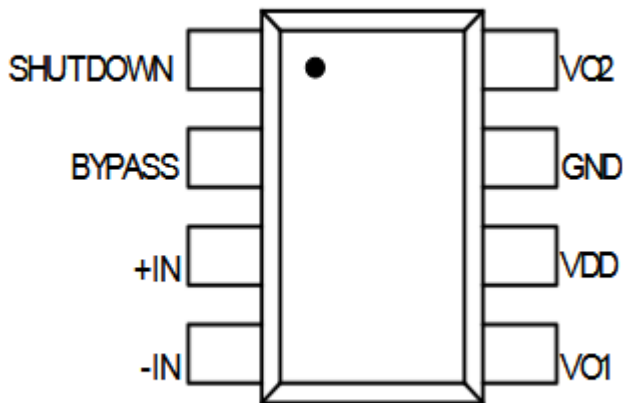
Audio Codec (ES8311)

The **ES8311** IC handles decoding and encoding. It is configured via I2C (address **0x18**) and transmits audio over the I2S bus. It processes incoming MIC signals (via ADC) and outputs audio signals to the amplifier (via DAC).



Audio Power Amplifier (FM8002E)

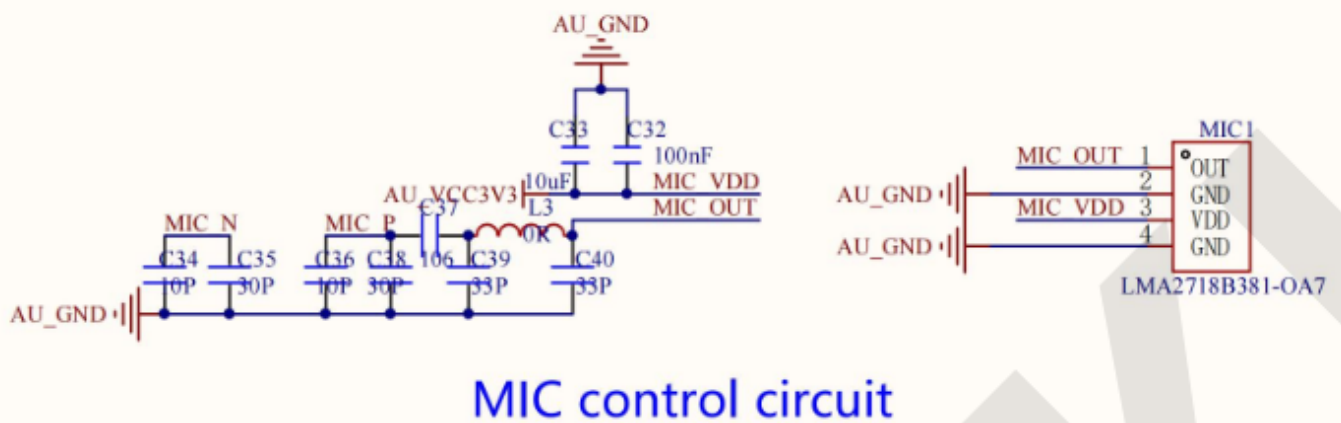
The **FM8002E** receives differential signals (OUTP/OUTN) from the codec and amplifies them for the speaker. At 5V, it can drive 1.5W (8 ohm load) or 2W (4 ohm load). The volume is controlled by gain adjustment resistors.



Microphone Control

A bottom-in MEMS silicon microphone is used for audio input. A CLC filtering circuit ensures clean signal transmission to the codec.

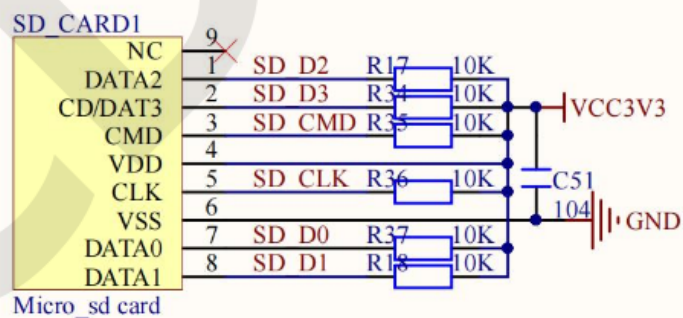
Hardware Identification Note: The PCB features a literal **MIC** marking alongside a small microphone icon/drawing indicating the exact location of the component.



6. Expansion & Interfaces

MicroSD Card Slot

Operates using the SDIO communication method rather than standard SPI, allowing for higher-speed storage access.

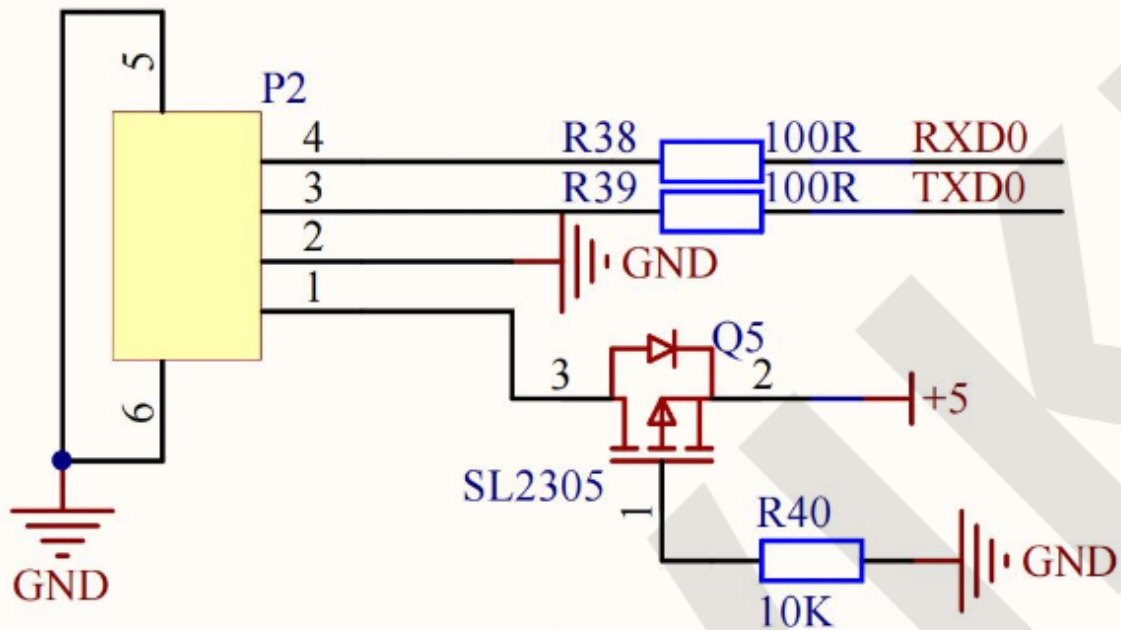


MicroSD card slot interface circuit

Serial Module (UART)

A 4-pin 1.25mm interface exposing RXD0 and TXD0. Features impedance balancing resistors and an anti-reverse field-effect transistor on the 5V line.

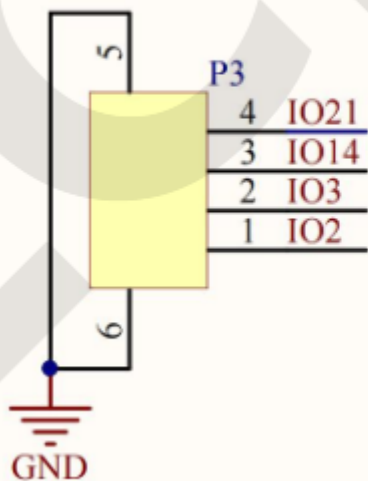
UART



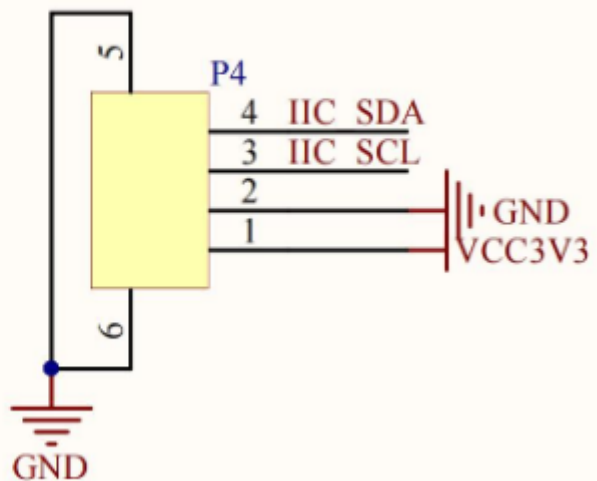
Expanded IO & I2C

Provides a 4-pin interface for general IO (IO21, IO14, IO3, IO2) which can double as an SPI bus, plus a dedicated 4-pin header for the I2C bus (IIC_SDA, IIC_SCL). The I2C pins cannot be used as standard GPIO.

Expanded IO



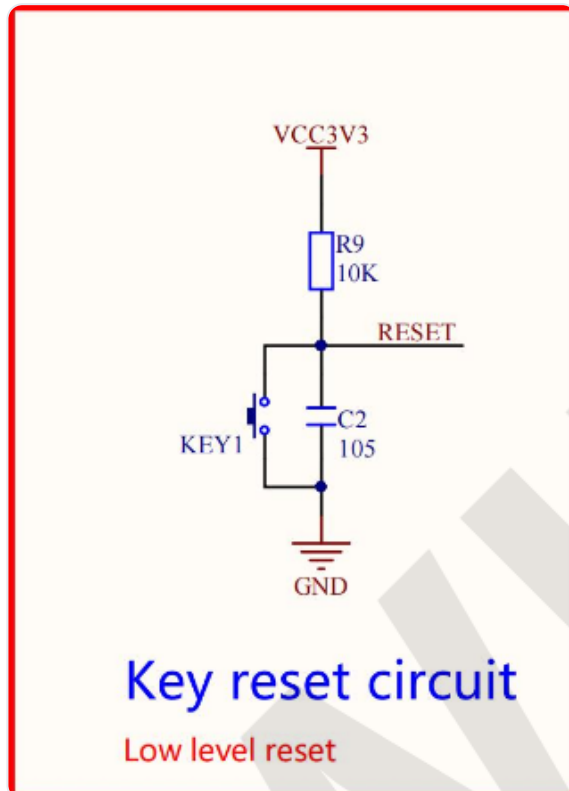
IIC



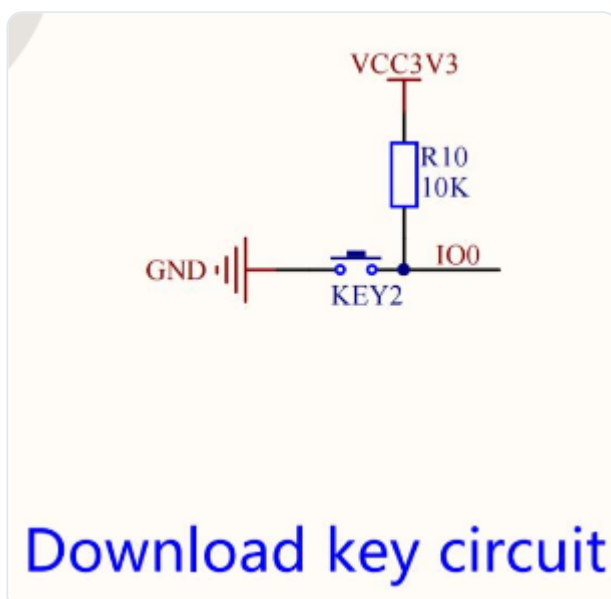
7. Controls & Indicators

Buttons

RESET Key (KEY1): Grounds the reset pin of the ESP32-S3. An RC circuit provides delay to ensure stable resets.



BOOT Key / Download Key (KEY2): Grounds IO0. Holding this while resetting places the ESP32-S3 into download mode. It can otherwise act as a general-purpose button.



RGB LED (WS2812B)

A built-in tri-color RGB LED utilizing an internal IC. Powered by 5V, it requires only a single IO pin to generate different color output based on timing waveforms.



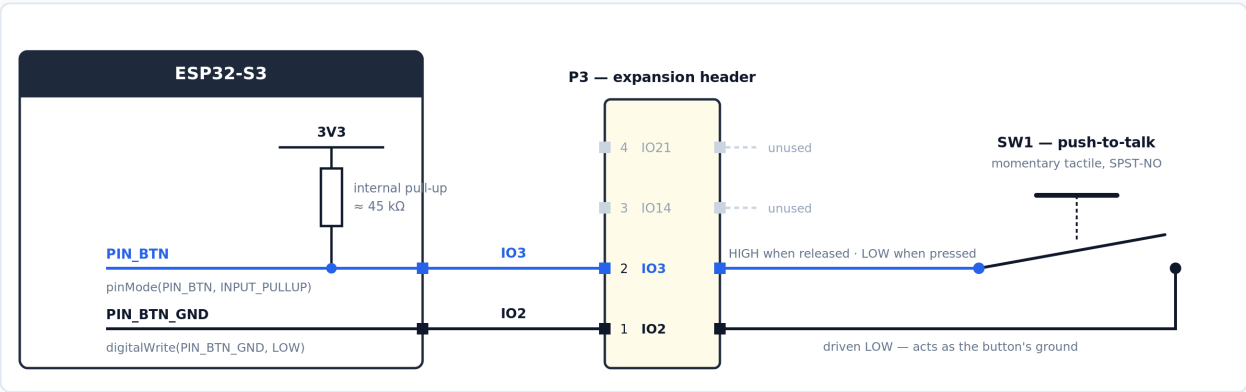
RGB three-color light control circuit

8. Push-to-Talk Button

Sections 1–7 above describe the stock module. This section documents wiring added for the Pip LLM Tutor project: an external momentary button the child holds while answering. It uses two pins of the **P3 expanded IO header** described in section 6, and needs no external components.

Push-to-Talk Button — Wiring to IO2 / IO3

Pip LLM Tutor · LCDWIKI 2.8" ESP32-S3 Display (ES3C28P) · P3 expansion header



Connections

The switch is wired directly between header pins 1 and 2. It is unpolarised, so either terminal may go to either pin.

P3 pin	Signal	Firmware constant	Mode
1	IO2	PIN_BTN_GND	OUTPUT , driven LOW — serves as the button's local ground
2	IO3	PIN_BTN	INPUT_PULLUP — reads HIGH when released, LOW when pressed
3	IO14	—	Unused
4	IO21	—	Unused

No pull-up resistor is fitted: IO3's internal pull-up (approximately 45 kΩ) is enabled in firmware, and IO2 is an ordinary GPIO held LOW so the button can occupy just two header pins instead of borrowing a ground.

Deep-sleep wake

The same button wakes the device. `EXT1` is armed to wake on IO3 going LOW, with IO2 held LOW through the RTC domain so it still acts as ground while the main GPIO drivers are off.

IO3 is a strapping pin with no default pull. The RTC peripheral domain must be kept powered during deep sleep, otherwise the internal pull-up is switched off, IO3 floats LOW and the module wakes the instant it sleeps. The cost is a few microamps of additional sleep current.

Fallback

If a unit's IO3 expansion line proves unreliable, set `PIN_BTN` to `0` in `pins.h` to fall back to the onboard BOOT key (section 7), which already grounds IO0. Note that IO43/IO44 (UART) cannot be used for this: TX idles HIGH and RX is held by the UART peripheral.